Vol 12 Chapter 2 — Chemical Bonding from Substrate

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 2 — Chemical Bonding from Substrate

Level 1 — one sentence

Chemical bonds — ionic, covalent, metallic, hydrogen, van der Waals — are atomic K-type couplings under multi-atom boundary conditions, with bond energies (1-1000 kJ/mol) and lengths (0.7-3 Å) following from substrate's natural electron K-type configurations.

Level 2 — graduate-physicist precision

2.1 Bond classification

- Ionic (NaCl, MgO): electron transfer; Coulomb-bound ions; 600-4000 kJ/mol
- Covalent (H_2, CH_4, diamond): electron sharing; molecular orbitals; 150-1000 kJ/mol
- Metallic (Fe, Cu, Al): delocalized electron sea; 80-800 kJ/mol
- Hydrogen (H_2O, DNA, proteins): dipole-dipole via H; 10-40 kJ/mol
- Van der Waals (noble gas dimers, lipids): induced dipole; 0.1-10 kJ/mol

2.2 Substrate-mechanism reading

Atomic K-types (Vol 5 Ch 6, Vol 9 Ch 3) under nuclear-skeleton boundary conditions interact via: - Electrostatic (ionic): substrate SO(2) weights → charges → Coulomb - Quantum overlap (covalent): substrate K-types share spatial structure - Delocalization (metallic): substrate K-types form continuous bands - Dipolar (hydrogen, vdW): substrate K-type asymmetry → moments

2.3 Bond length and energy

Bond length $\sim a_0$ (Bohr radius) up to a few $\times\, a_0$. Bond energy $\sim$ eV per bond = 96 kJ/mol per eV.

Examples: - H-H: 0.74 Å, 432 kJ/mol - C-C: 1.54 Å, 348 kJ/mol
- O=O: 1.21 Å, 498 kJ/mol - N≡N: 1.10 Å, 945 kJ/mol

2.4 Hybridization (S-tier per Grace INV-5124, 2026-05-24)

Atomic orbitals mix into hybrid orbitals for bonding: - sp^3: tetrahedral (CH_4, diamond) — 4 hybrid orbitals - sp^2: trigonal planar (graphene, benzene) — 3 hybrid orbitals - sp: linear (acetylene) — 2 hybrid orbitals - sp^3d, sp^3d^2: 5- and 6-coordinate

BST-primary pattern (partial, S-tier per Grace catalog): sp = 2 = rank ⍰; $sp^2 = 3 = N_c$ ⍰; $sp^3 = 4$ = not a BST primary ×. The pattern breaks at sp^3, so "substrate K-type reorganization" framing for hybridization is ASSERTED, not derived — demoted from I-tier to S-tier structural observation per Grace INV-5124 Sunday 2026-05-24. Promotion path: multi-week Lyra mechanism work (analogous to amino-acid #306) deriving substrate-natural hybridization from K-type representation theory, OR honest documentation that hybridization is a chemistry artifact without BST mechanism.

2.5 K-audit anchors

- Vol 5 Ch 6: hydrogen atom (single-atom K-type)
- Vol 9 Ch 3: band structure (multi-atom)

Level 3 — 5th-grader accessibility

Chemical bonds hold atoms together. Ionic (NaCl): electron transferred, ions attract. Covalent (H_2): electrons shared. Metallic (Fe): delocalized electron sea. Hydrogen (water): dipole-dipole via H. Van der Waals: weak induced-dipole. Bond strengths: 0.1 kJ/mol (vdW) to 4000 kJ/mol (strong ionic). In BST: bonds are substrate K-type couplings under nuclear-skeleton boundary conditions.

What comes next

Chapter 3 develops molecular orbital theory.

Where to look this up

- Pauling, *Nature of the Chemical Bond*
- BST: Vol 5 Ch 6; Vol 9 Ch 3